4

Classes of Calibres, Using Σ-products

We investigate the extent of classes of calibres of spaces; i.e., we characterize those classes of infinite cardinals which arise, for some space X, as the class of calibres of X. (For technical reasons it turns out to be more convenient to determine classes of non-calibres than classes of calibres. The resulting description (Corollary 4.7) is satisfactory in that those restrictions which are clearly necessary on the basis of the preceding chapters turn out to be sufficient.) The spaces we define with calibre properties prescribed in advance are products of Σ-products of the generalized Cantor powers 2^I. These are dense subspaces of the powers 2^I, and accordingly their Souslin number is equal to ω^+ and they are not compact; a consequence of (the proof of) Corollary 4.8 is that if for a class $\mathbf{T}$ of infinite cardinals there is a Hausdorff space X such that X does not have calibre α if and only if $\alpha \in \mathbf{T}$, then X may be chosen to be a completely regular, Hausdorff space such that $S(X) = \omega^+$.

The question of the extent of calibres of compact spaces is more delicate and is treated in detail in Chapter 5.

We have seen in Chapter 3 that for regular cardinals the calibre properties of a product space X_I are determined by the products X_J with $J \in \mathscr{P}^*_\omega(I)$. An argument given by Gerlits shows (Theorem 4.4) that the analogous statement fails for every (infinite) singular cardinal.

Definition. Let κ be an infinite cardinal, $\{X_i : i \in I\}$ a set of sets and

$$p = \langle p_i : i \in I \rangle \in X_I = \prod_{i \in I} X_i.$$

For $x \in X_I$ the *exceptional set of* x (with respect to p), denoted $E_p(x)$, is the set

$$E_p(x) = \{i \in I : x_i \neq p_i\};$$

and the *κ-Σ-product of* X_I *based at* p, denoted $\Sigma_\kappa(p, X_I)$, is the set

$$\{x \in X_I : |E_p(x)| < \kappa\}.$$

The set $\Sigma_\kappa(p, X_I)$ is a *generalized Σ-product*. When ambiguity is unlikely we write $E(x)$ in place of $E_p(x)$ and $\Sigma_\kappa(p)$ in place of $\Sigma_\kappa(p, X_I)$.

If $\lambda \geq \omega$ and p is the point of $\{0,1\}^\lambda$ defined by

$$p_\xi = 0 \quad \text{for } \xi < \lambda,$$

we write $\Sigma_\kappa\{0,1\}^\lambda$ in place of $\Sigma_\kappa(p, \{0,1\}^\lambda)$.

We note that if $\kappa > |I|$ then $\Sigma_\kappa(p) = X_I$ for all $p \in X_I$.

The basic projection properties of generalized Σ-products are given in this simple result.

4.1 *Theorem*. Let $\kappa \geq \omega$, let $\{X_i : i \in I\}$ be a set of sets, $p \in X_I$ and $Y = \Sigma_\kappa(p)$. Then

(a) $\pi_J[Y] = X_J$ for every non-empty $J \subset I$ with $|J| < \kappa$; and

(b) if the sets X_i are spaces then for every compact $K_J \subset X_J$ with J non-empty, $J \subset I$ and $|J| < \kappa$ there is compact $K \subset Y$ such that $\pi_J[K] = K_J$.

Proof. (a) For $x \in X_J$ we set

$$y = \langle x, p_{I \setminus J} \rangle \in X_J \times X_{I \setminus J} = X_I;$$

it is clear that $y \in Y$ and $\pi(y) = x$.

(b) We set $K = K_J \times \{p_{I \setminus J}\}$. Then K is a compact subspace of Y such that $\pi_J[K] = K_J$.

It follows from Theorem 4.1 that certain generalized Σ-products satisfy the hypotheses, and hence the conclusions, of Theorems 3.3, 3.5 and 3.8.

We have noted already that an infinite Hausdorff space does not have calibre α when $\mathrm{cf}(\alpha) = \omega$. Since there are arbitrarily large cardinals α such that $\mathrm{cf}(\alpha) = \omega$, for every infinite space X the class of cardinals α such that X does not have calibre α is a proper class. On the other hand it follows from Theorem 2.6 that every regular cardinal α satisfying $\alpha > d(X)$ is a calibre for X; thus for every space X the class

$$\{\alpha : \alpha \text{ is a regular cardinal and } X \text{ does not have calibre } \alpha\}$$

is a set. Thus it is more natural, given a space X, to specify those infinite cardinals α which are not calibres for X than to identify those that are.

We do this now for certain generalized Σ-products.

4.2 *Theorem.* Let α, κ and λ be infinite cardinals and $X = \Sigma_\kappa\{0,1\}^\lambda$. The following statements are equivalent.

(a) $\mathrm{cf}(\alpha) = \omega$ or $\kappa \leq \alpha \leq \lambda$ or $\kappa \leq \mathrm{cf}(\alpha) \leq \lambda$;

(b) X does not have calibre α.

Proof. (a)$\Rightarrow$(b). We have $|X| \geq \omega$; from Theorems 2.10 and 2.3(a) it follows that X does not have calibre α if $\mathrm{cf}(\alpha) = \omega$.

We note next that X does not have calibre (λ, κ). Indeed let

$$U^\eta = \{x \in X : x_\eta = 1\} \quad \text{for } \eta < \lambda;$$

then $\{U^\eta : \eta < \lambda\} \subset \mathscr{T}^*(X)$, and $\cap_{\eta \in A} U^\eta = \varnothing$ for $A \in [\lambda]^\kappa$.

Now if either $\kappa \leq \alpha \leq \lambda$ or $\kappa \leq \mathrm{cf}(\alpha) \leq \lambda$, then (from Theorem 2.1(a)) either X does not have calibre α or X does not have calibre $\mathrm{cf}(\alpha)$. It follows from Theorem 2.3(a) that X does not have calibre α.

(b)$\Rightarrow$(a). We assume $\mathrm{cf}(\alpha) > \omega$ and we consider four cases.

Case 1. $\kappa > \lambda$. Then $X = \{0,1\}^\lambda$ and X has calibre α by Theorem 3.18.

Case 2. $\kappa \leq \lambda$ and $\alpha < \kappa$. Let $\{U^\xi : \xi < \alpha\}$ be a set of basic subsets of X and set $J = \cup_{\xi < \alpha} R(U^\xi)$. Since $\{0,1\}^J$ has calibre α (by Theorem 3.18) there are $A \in [\alpha]^\alpha$ and $p \in \{0,1\}^J$ such that

$$p \in \bigcap_{\xi \in A} \pi_J[U^\xi].$$

We define $x \in \{0,1\}^\lambda$ by

$$\begin{aligned} x_\eta &= p_\eta \quad \text{if } \eta \in J \\ &= 0 \quad \text{if } \eta \in \lambda \backslash J; \end{aligned}$$

then $x \in \cap_{\xi \in A} U^\xi$, and from $|J| \leq \alpha < \kappa$ we have $x \in X$.

Case 3. $\kappa \leq \lambda$ and $\lambda < \mathrm{cf}(\alpha)$. There is a pseudobase $\mathscr{B}$ for X such that $|\mathscr{B}| = \lambda$. Let $\{U^\xi : \xi < \alpha\} \subset \mathscr{T}^*(X)$ and set

$$A(B) = \{\xi < \alpha : B \subset U^\xi\} \quad \text{for } B \in \mathscr{B}.$$

Then $\cap_{\xi \in A(B)} U^\xi \supset B \neq \varnothing$ for $B \in \mathscr{B}$, and since $\lambda < \mathrm{cf}(\alpha)$ there is $\bar{B} \in \mathscr{B}$ such that $|A(\bar{B})| = \alpha$.

Case 4. $\kappa \leq \lambda$ and $\mathrm{cf}(\alpha) < \kappa$ and $\lambda < \alpha$. It follows from Case 2 (applied to $\mathrm{cf}(\alpha)$ in place of α) and X has calibre $\mathrm{cf}(\alpha)$.

Since $w(\{0,1\}^\lambda) = \lambda$ we have $\pi w(X) < \alpha$ and hence from Theorem 2.3(a′) the space X has calibre α.

In each of the four cases the space X has calibre α.

The proof is complete.

In Theorem 8.11 we determine those (infinite) cardinals α for which the spaces $\Sigma_\kappa\{0,1\}^\lambda$ have compact-calibre α and are pseudo-(α,α)-compact.

We see next that in Theorem 2.3(a′) the condition $\pi w(X) < \alpha$ cannot be omitted.

4.3 *Theorem.* Let α be a singular cardinal such that $\mathrm{cf}(\alpha) > \omega$. There is a completely regular, Hausdorff space X such that X has calibre $\mathrm{cf}(\alpha)$, X does not have calibre α, and $\pi w(X) = \alpha$.

Proof. We set $X = \Sigma_{(\mathrm{cf}(\alpha))^+}\{0,1\}^\alpha$. It is immediate from Theorem 4.2 that X has the required properties.

We saw in Corollary 3.7(a) that if α is an uncountable regular cardinal and $\{X_i : i \in I\}$ a set of spaces such that X_i has calibre α for all $i \in I$, then X_I has calibre α. We see next that the analogous statement fails for singular cardinals α. Theorem 4.4 supplements and complements Corollary 3.11(i), which treats products of spaces with calibre (α,β) with α singular and $\alpha > \beta$.

4.4 *Theorem.* Let α be an infinite singular cardinal. There is a set $\{X_i : i \in I\}$ of completely regular, Hausdorff spaces such that

X_i has calibre α for all $i \in I$, and
X_I does not have calibre α.

Proof. If $\mathrm{cf}(\alpha) = \omega$ we establish the statement by setting $I = \omega$ and $X_n = \{0,1\}$ for $n < \omega$. We assume in what follows that $\mathrm{cf}(\alpha) > \omega$.

There is a set $\{A_\sigma : \sigma < \mathrm{cf}(\alpha)\}$ of sets such that

$$\mathrm{cf}(\alpha) < |A_\sigma| < \alpha \text{ for } \sigma < \mathrm{cf}(\alpha),$$
$$A_{\sigma'} \cap A_\sigma = \emptyset \text{ for } \sigma' < \sigma < \mathrm{cf}(\alpha), \text{ and}$$
$$\bigcup_{\sigma < \mathrm{cf}(\alpha)} A_\sigma = \alpha.$$

We set

$$X_\sigma = \Sigma_{(\mathrm{cf}(\alpha))^+}\{0,1\}^{A_\sigma} \text{ for } \sigma < \mathrm{cf}(\alpha), \text{ and}$$

$$X = \prod_{\sigma < \mathrm{cf}(\alpha)} X_\sigma.$$

From Theorem 4.2 (with κ and λ replaced by $(\mathrm{cf}(\alpha))^+$ and $|A_\sigma|$, respectively) it follows that X_σ has calibre α. Then from Theorem 2.2(a) it follows that X_J has calibre α for all $J \in \mathscr{P}^*_\omega(I)$.

We are to show that X does not have calibre α. We show that in fact X does not have calibre $(\alpha, (\mathrm{cf}(\alpha))^+)$.

We have

$$X \subset \prod_{\sigma < \mathrm{cf}(\alpha))^+} \{0,1\}^{A_\sigma} = \{0,1\}^\alpha$$

and in fact

$$X = \Sigma_{(\mathrm{cf}(\alpha))^+}\{0,1\}^\alpha.$$

We set

$$U_\xi = \{x \in X : x_\xi = 1\} \text{ for } \xi < \alpha;$$

then $\{U_\xi : \xi < \alpha\} \subset \mathscr{T}^*(X)$, and

$$\bigcap_{\xi \in A} U_\xi = \varnothing \quad \text{for } A \in [\alpha]^{(\mathrm{cf}(\alpha))^+},$$

as required.

We note that for $\mathrm{cf}(\alpha) > \omega$ the space X_I of Theorem 4.1 does have calibre $\mathrm{cf}(\alpha)$.

We give now a consequence of Theorem 4.2, and a theorem which allows us to determine those classes of infinite cardinal numbers which can arise as the class of non-calibres of a space.

4.5 *Corollary*. Let λ and α be infinite cardinals and $X = \Sigma_\lambda\{0,1\}^\lambda$. Then X has calibre α if and only if $\alpha \neq \lambda$, $\mathrm{cf}(\alpha) \neq \lambda$, and $\mathrm{cf}(\alpha) \neq \omega$.

4.6 *Theorem*. Let $\mathbf{S}$ be a non-empty set of infinite cardinal numbers, $X(\lambda) = \Sigma_\lambda\{0,1\}^\lambda$ for $\lambda \in \mathbf{S}$ and $X(\mathbf{S}) = \prod_{\lambda \in \mathbf{S}} X(\lambda)$.

For $\alpha \geq \omega$, the following statements are equivalent.

(a) α is not a calibre for $X(\mathbf{S})$;

(b) $\alpha \in \mathbf{S}$ or $\mathrm{cf}(\alpha) \in \mathbf{S}$ or $\mathrm{cf}(\alpha) = \omega$.

Proof. (a) $\Rightarrow$ (b). We suppose that $\alpha \notin \mathbf{S}$, $\mathrm{cf}(\alpha) \notin \mathbf{S}$ and $\mathrm{cf}(\alpha) > \omega$. It follows from Theorem 4.2 that $X(\lambda)$ has calibre α for all $\lambda \in \mathbf{S}$.

We show that $X(\mathbf{S})$ has calibre α. If α is regular this follows from Corollary 3.7(a). We assume in what follows that α is singular.

Let $\{\alpha_\sigma : \sigma < \mathrm{cf}(\alpha)\}$ be a set of cardinal numbers such that

α_σ is regular for $\sigma < \mathrm{cf}(\alpha)$,

$\mathrm{cf}(\alpha) < \alpha_{\sigma'} < \alpha_\sigma < \alpha$ for $\sigma' < \sigma < \mathrm{cf}(\alpha)$ and

$$\sum_{\sigma < \mathrm{cf}(\alpha)} \alpha_\sigma = \alpha.$$

We claim that if $J \in \mathscr{P}^*_\omega(\mathbf{S})$ then X_J has calibre α and calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$. That X_J has calibre α follows from Theorem 2.2(a); to prove that X_J has calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ we consider two cases.

Case 1. $\alpha_\sigma \notin J$. Then $X(\lambda)$ has calibre α_σ for all $\lambda \in J$. It follows that X_J has calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$.

Case 2. $\alpha_\sigma \in J$. If $J \backslash \{\alpha_\sigma\} = \varnothing$ then X_J, which is $X(\alpha_\sigma)$, has calibre $\alpha_{\sigma+1}$ and hence calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$. If $J \backslash \{\alpha_\sigma\} \neq \varnothing$ then, since $X_{J \backslash \{\alpha_\sigma\}}$ has calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ (by Case 1) and $X(\alpha_\sigma)$ has calibre $\alpha_{\sigma+1}$, the space X_J, which is $X_{J \backslash \{\alpha_\sigma\}} \times X(\alpha_\sigma)$, has calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$.

The proof of the claim is complete. It follows from Corollary 3.11(a) that $X(\mathbf{S})$ has calibre α, as required.

(b) $\Rightarrow$ (a). From Corollary 4.5 it follows that there is $\lambda \in \mathbf{S}$ such that $X(\lambda)$ does not have calibre α. It is then clear that $X(\mathbf{S})$ does not have calibre α.

4.7 *Corollary*. Let $\mathbf{T}$ be a class of infinite cardinal numbers. Then statements (a) and (b) are equivalent.

(a) There is an infinite Hausdorff space X such that X does not have calibre α if and only if $\alpha \in \mathbf{T}$.

(b) (i) $\omega \in \mathbf{T}$,

(ii) if $\mathrm{cf}(\alpha) \in \mathbf{T}$ then $\alpha \in \mathbf{T}$,

(iii) $\{\mathrm{cf}(\alpha) : \alpha \in \mathbf{T}\}$ is a set, and

(iv) if κ is a regular cardinal and $\kappa \notin \mathbf{T}$, then $\{\alpha \in \mathbf{T} : \mathrm{cf}(\alpha) = \kappa\}$ is a set.

Proof. (a) $\Rightarrow$ (b). (i) follows from Corollary 2.10.

(ii) follows from Theorem 2.3(a).

(iii) If α is an infinite cardinal and $\mathrm{cf}(\alpha) > d(X)$, then it follows from Theorem 2.6 that X has calibre α. Hence

$$\{\mathrm{cf}(\alpha) : \alpha \in \mathbf{T}\} \subset (d(X))^+.$$

(iv) From Theorem 2.3(a′) we have

$$\{\alpha \in \mathbf{T} : \mathrm{cf}(\alpha) = \kappa\} \subset (\pi w(X))^+ \quad \text{for } \kappa \notin \mathbf{T}.$$

(b) ⇒ (a). For $\kappa \in \{\mathrm{cf}(\alpha) : \alpha \in \mathbf{T}\}$ we set

$$\begin{aligned} \mathbf{S}(\kappa) &= \{\kappa\} && \text{if } \kappa \in \mathbf{T} \\ &= \{\alpha \in \mathbf{T} : \mathrm{cf}(\alpha) = \kappa\} && \text{if } \kappa \notin \mathbf{T}, \end{aligned}$$

we set

$$\mathbf{S} = \cup \{\mathbf{S}(\kappa) : \kappa \in \{\mathrm{cf}(\alpha) : \alpha \in \mathbf{T}\}\},$$

we note that $\mathbf{S}$ is a set and $\mathbf{S} \subset \mathbf{T}$, and we set

$$X(\lambda) = \Sigma_\lambda \{0, 1\}^\lambda \text{ for } \lambda \in \mathbf{S}, \text{ and}$$

$$X = \prod_{\lambda \in \mathbf{S}} X(\lambda).$$

We show first that if α is an infinite cardinal and X does not have calibre α then $\alpha \in \mathbf{T}$. From Theorem 4.6 we have $\alpha \in \mathbf{S}$ or $\mathrm{cf}(\alpha) \in \mathbf{S}$ or $\mathrm{cf}(\alpha) = \omega$. Since $\mathbf{S} \subset \mathbf{T}$, from (ii) we have $\alpha \in \mathbf{T}$ if $\alpha \in \mathbf{S}$ or $\mathrm{cf}(\alpha) \in \mathbf{S}$; and from (i) and (ii) we have $\alpha \in \mathbf{T}$ if $\mathrm{cf}(\alpha) = \omega$.

We show next that if $\alpha \in \mathbf{T}$ then X does not have calibre α. If $\mathrm{cf}(\alpha) \in \mathbf{T}$ then

$$\mathrm{cf}(\alpha) \in \mathbf{S}(\mathrm{cf}(\alpha)) \subset \mathbf{S},$$

and if $\mathrm{cf}(\alpha) \notin \mathbf{T}$ then $\alpha \in \mathbf{S}(\mathrm{cf}(\alpha)) \subset \mathbf{S}$; it follows from Theorem 4.6 that X does not have calibre α.

4.8 *Remark.* We note that the space X defined in the proof of the implication (b) ⇒ (a) in Corollary 4.7 is a dense subspace of a generalized Cantor power 2^I. The space X is then a completely regular, Hausdorff space, and from Theorem 3.25 and Remark 2.6(b) it follows that $S(X) = \omega^+$. This shows that if for a class $\mathbf{T}$ of infinite cardinal numbers there is a Hausdorff space X such that X does not have calibre α if and only if $\alpha \in \mathbf{T}$, then X may be chosen to be a completely regular, Hausdorff space such that $S(X) = \omega^+$.

Notes for Chapter 4

The term Σ-product was introduced by Corson (1959) and applied to spaces denoted (in our notation) $\Sigma_{\omega^+}(p, X_I)$.

Theorem 4.2 is noted by Comfort (1980, Theorem 4.1(a)).

Comfort (1971, p. 175) raised the question of 'characterizing those collections of cardinal numbers which can arise as the totality of calibres possessed by some space'. Theorem 4.6, responding to this question, is from Shelah (1977, Theorem 3.3), though the spaces defined by Shelah are not regular spaces. Our proof of Theorem 4.6, systematically exploiting the calibre properties of the generalized Σ-products $\Sigma_\kappa\{0,1\}^\lambda$ (Theorem 4.2), closely parallels the treatment (obtained independently) of Broverman, Ginsburg, Kunen & Tall (1978); these authors consider in addition a number of properties (e.g., normality and 'Baireness'), and their behavior under special set-theoretic assumptions, of spaces of the form

$$\Sigma(\mathscr{I}) = \{x \in \{0,1\}^{(\omega^+)} : \{\xi < \omega^+ : x_\xi = 1\} \in \mathscr{I}\}$$

with $\mathscr{I}$ a σ-ideal in ω^+.

The example of Theorem 4.4 was communicated in conversation by J. Gerlits (February, 1978).